

E0X Calculating Xseg and Xmm from RTD data

A sample of the tracer hythane was injected as a pulse into a reactor and the exit concentration was measured as a function of time. The data is given below.

1. Construct a curve showing $C(t)$.
2. Construct the $E(t)$ curve.
3. Construct the $F(t)$ curve.
4. Compute X_{CSTR} , X_{PFR} , X_{seg} , and X_{mm} for the following elementary liquid phase reactions:
 - a. $A \rightarrow B$ with $k = 0.1 \text{ min}^{-1}$
 - b. $2A \rightarrow B$ with $k = 0.1 \text{ m}^3/\text{kmol}/\text{min}$ and $C_{A0} = 1.0 \text{ kmol}/\text{m}^3$

t (min)	0	0.5	1	2	3	4	5	6	7	8	9	10	12	14
C (g/m ³)	0	0.6	1.4	5	8	10	8	6	4	3	2.2	1.6	0.6	0

consider $2A \rightarrow B$ $-r_A = k C_A^2 = k C_{A0}^2 (1-x)^2$

CSTR $V = \frac{F_{A0} X}{-r_A}$

$$V = \frac{F_{A0} X}{-r_A} = \frac{F_{A0} X}{k C_{A0}^2 (1-x)^2} = \frac{C_{A0} v_0 X}{k C_{A0}^2 (1-x)^2} = \frac{v_0 X}{k C_{A0} (1-x)^2}$$

$$\gamma = \frac{X}{k C_{A0} (1-x)^2}, \quad \gamma k C_{A0} = \frac{X}{(1-x)^2}$$

$\uparrow$ $\uparrow$ $\uparrow$
 t_m $0.1 \frac{\text{m}^3}{\text{kmol} \cdot \text{min}}$ $= 1 \frac{\text{kmol}}{\text{m}^3}$

$\approx 5.11 \text{ min}$

$$X_{\text{CSTR}} = 0.27$$

PFR: $\frac{dX}{dV} = \frac{-r_A}{F_{A0}} = \frac{k C_{A0}^2 (1-X)^2}{F_{A0}} = \frac{k C_{A0}^2 (1-X)^2}{C_{A0} v_0}$

$$\frac{dX}{dV} = \frac{k C_{A0} (1-X)^2}{v_0}$$

$$\frac{dX}{d\tau} = k C_{A0} (1-X)^2 \quad \int_0^X \frac{dX}{(1-X)^2} = k C_{A0} \tau$$

$0.1 \frac{\text{kmol}}{\text{m}^3 \cdot \text{min}}$
 $\downarrow$
 611 min
 $\uparrow$
 $1 \frac{\text{kmol}}{\text{m}^3}$

$$k C_{A0} \tau = \frac{X}{1-X}$$

$$X_{\text{PFR}} = 0.34$$

Segregation model $\bar{X} = \int_0^\infty X(t) E(t) dt$

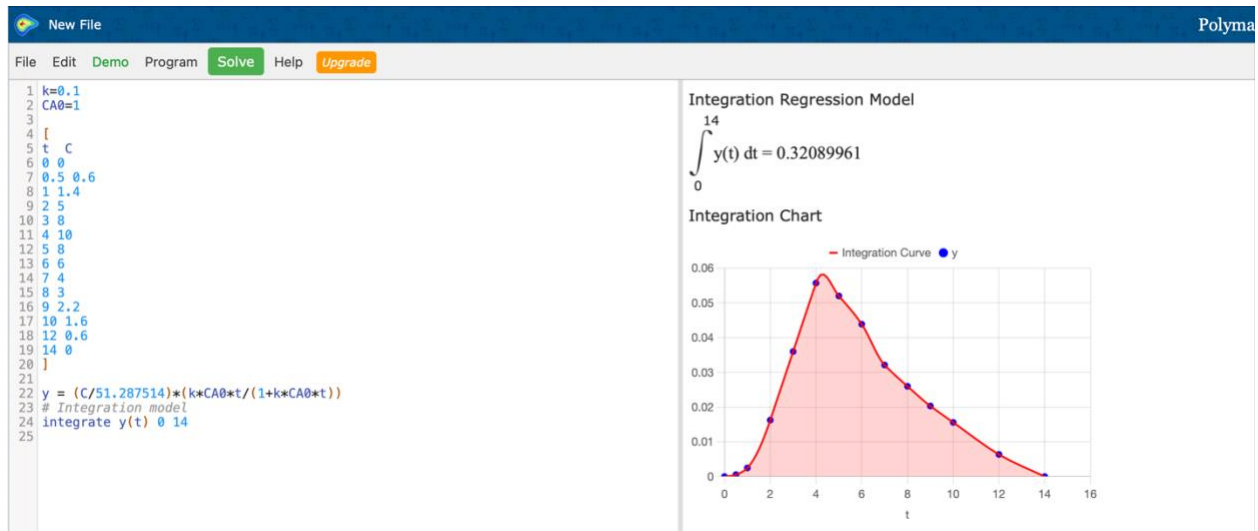
comes from the batch reactor mole balance

to get $X(t)$: write batch reactor mole balance

$$\frac{dx}{dt} = \frac{-r_A}{C_{A0}} = \frac{k C_{A0}^2 (1-x)^2}{C_{A0}} = k C_{A0} (1-x)^2$$

$$\int_0^x \frac{dx}{(1-x)^2} = k C_{A0} t \rightarrow \frac{x}{1-x} = k C_{A0} t$$

$$\text{Solve for } x: \quad x = \frac{k C_{A0} t}{1 + k C_{A0} t} = x(t)$$



$$\int_0^{\infty} X(t) E(t) dt = 0.32, \text{ determined w/ Polymath Plus}$$

$$X_{\text{seq}} = 0.32$$

maximum mixedness model

$$\frac{dX}{dz} = \frac{-r_A}{C_{A0}} - \frac{E(\bar{T}-z)}{1-F(\bar{T}-z)} X$$

$$z = t \quad \bar{T} = 14 \text{ min}$$

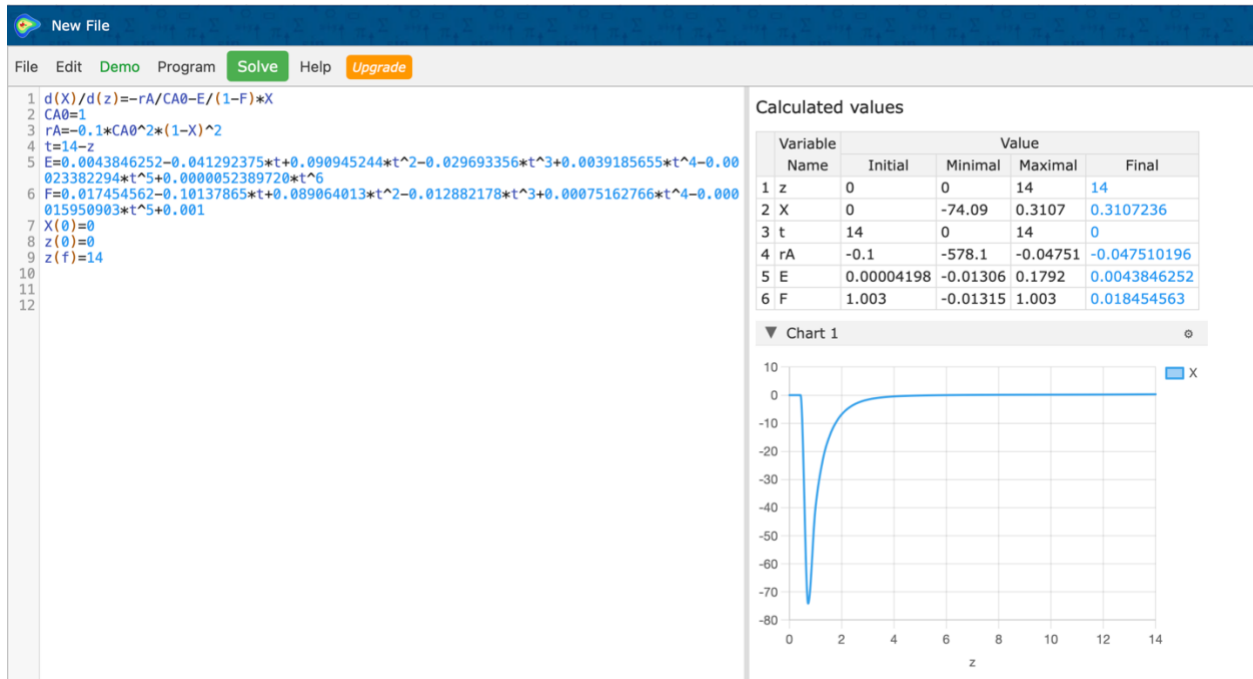
$$-r_A = k C_{A0}^2 (1-X)^2$$

$$C_{A0} = 1 \text{ kmol/m}^3$$

I am going to create best fit functions for $E(t)$

& $F(t)$ for the sake of ease

$$X_{mm} = 0.31$$



polynomials fit & data integrated using
Polymath Plus

Summary:

<u>model</u>	<u>X</u>
CSTR	0.27
PFR	0.34
	0.32
Segregation	0.32
mm	0.31